



2019

**Preliminary
Assessment
Task 2**

Preliminary Mathematics Extension

General Instructions

- Reading Time – 2 minutes
- Working Time – 55 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 5 – 7, show relevant mathematical reasoning and/or calculations

Total marks – 40

Section I – 4 marks (pages 2 – 4)

- Attempt Questions 1 – 4
- Allow about 7 minutes for this section

Section II – 36 marks (pages 5 – 7)

- Attempt Questions 5 – 7
- Allow about 48 minutes for this section

NAME: _____

TEACHER: _____

Question	1 – 4	5	6	7	Total
Further Trig		/8	/12	/11	/31
Further Algebra and Functions	/4	/5			/9
	/4	/13	/12	/11	/40

Section I

4 marks

Attempt Questions 1- 4

Allow about 7 minutes for this section

Use the multiple-choice answer sheet for Questions 1- 4

1. What is the solution of the inequation $|2x - 1| < 7$?

(A) $\frac{1}{2} < x < 4$

(B) $-4 < x < 3$

(C) $-\frac{1}{2} < x < 4$

(D) $-3 < x < 4$

2. What is the solution of the inequation $x(x-3)^2(x+3)^3 \leq 0$?

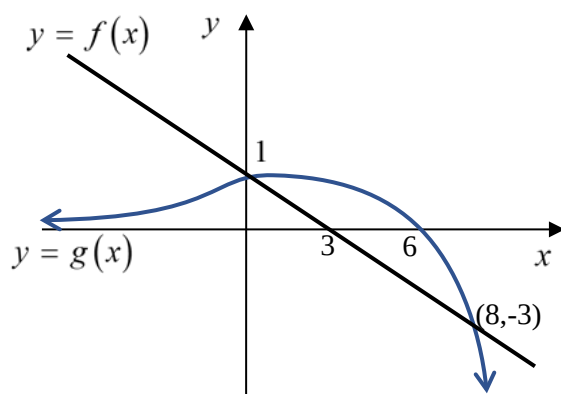
(A) $-3 \leq x \leq 0, \quad x \geq 3$

(B) $-3 \leq x \leq 0, \quad x = 3$

(C) $0 \leq x \leq 3, \quad x \leq -3$

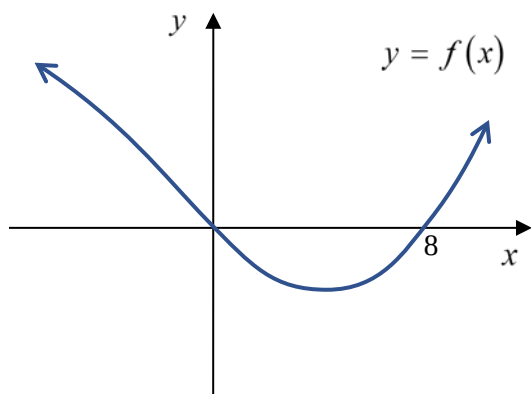
(D) $0 \leq x \leq 3, \quad x = -3$

3. The graphs of the functions $y = f(x)$ and $y = g(x)$ are shown on the same axes.

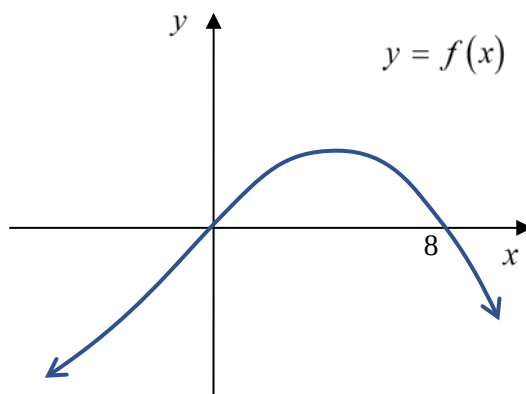


Which of the following is a possible graph of $y = f(x) - g(x)$

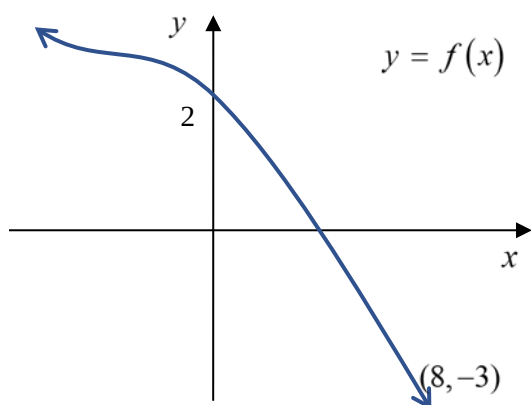
(A)



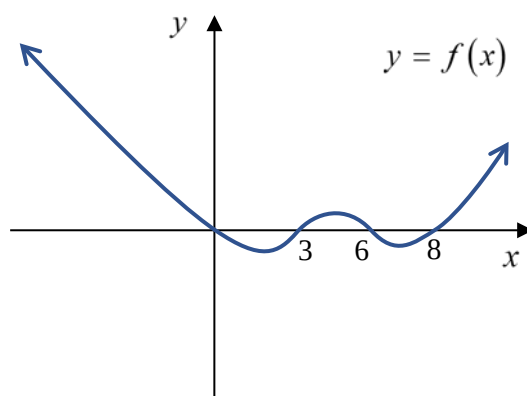
(B)



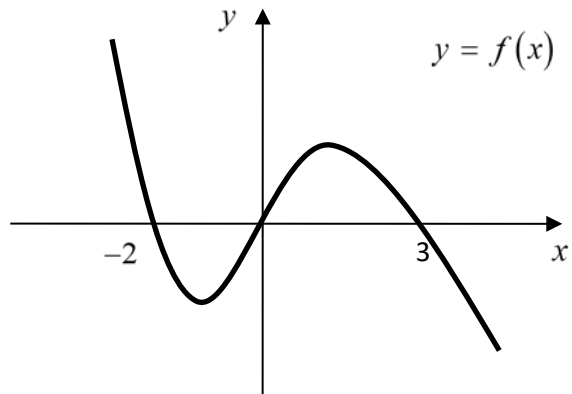
(C)



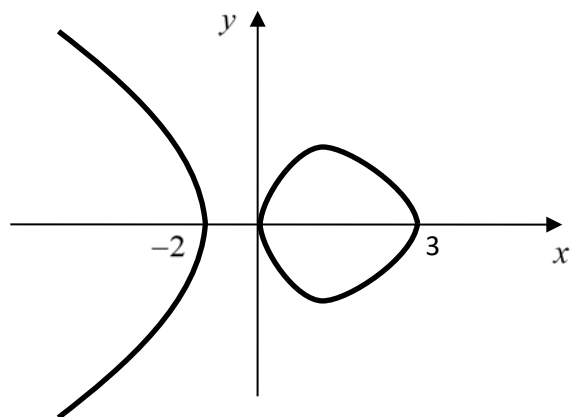
(D)



4. The graph of the function $y = f(x)$ is shown.



A second graph is obtained from the function $y = f(x)$



Which equation best represents the second graph?

- (A) $y = \sqrt{f(x)}$
- (B) $y = f(\sqrt{x})$
- (C) $y^2 = f(x)$
- (D) $|y| = f(|x|)$

Section II

36 marks

Attempt Questions 5-7

Allow about 48 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In questions 5-7, your responses should include relevant mathematical reasoning and/or calculations.

Question 5 (13 marks) Use a SEPARATE writing booklet.

(a) If $f(x) = \frac{x}{x-3}$, find $g(x)$ such that $g(x) = f^{-1}(x)$ 2

(b) Solve $\frac{2x}{x-3} \leq x$. 3

(c) By using a compound angle formula, determine the exact value of $\tan 75^\circ$. 2

(d) Let $5 \cos x - \sqrt{11} \sin x \equiv R \cos(x + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 360^\circ$.

(i) By expanding the right-hand side and equating coefficients, 3
show that $R = 6$ and $\alpha = 33^\circ 33'$ to the nearest minute.

(ii) Hence, find the range of $y = 6 + 5 \cos x - \sqrt{11} \sin x$. 1

(e) Show that $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$. 2

End of Question 5

Question 6 (12 marks) Use a SEPARATE writing booklet.

(a) Solve $\sin 2x - \sin x = 0$ for $-\pi \leq x \leq \pi$. **3**

(b) By letting $t = \tan \frac{\theta}{2}$, solve: **3**

$$\frac{\cos \theta + \sin \theta - 1}{\cos \theta - \sin \theta + 1} = 3 \quad \text{for } 0 \leq \theta \leq 2\pi$$

(c) Consider the parametric equations $y = \cos 2t - 1$ and $x = \cos t$.

(i) Determine the Cartesian equation in x and y . **2**

(ii) What are the restrictions on x and y ? **2**

(iii) Sketch the curve defined by $y = \cos 2t - 1$ and $x = \cos t$. **2**

End of Question 6

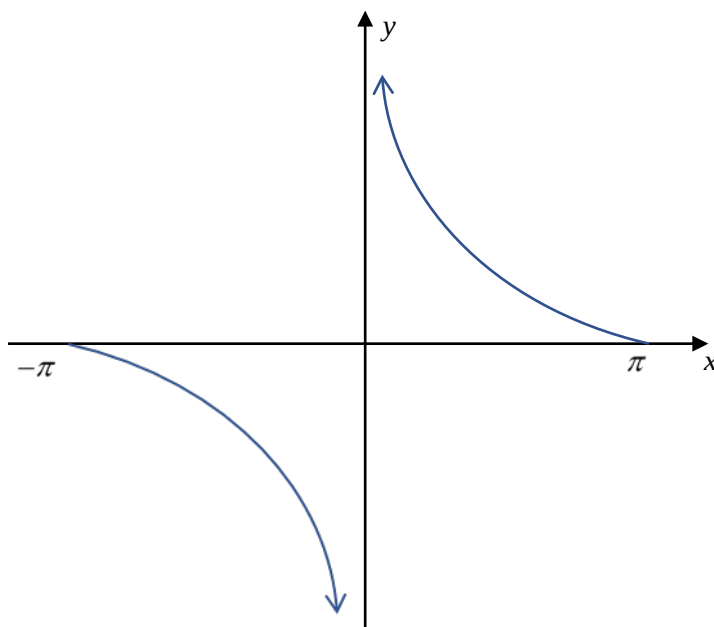
Question 7 (11 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that $\cos(\alpha - \beta) + \cos(\alpha + \beta) = 2 \cos \alpha \cos \beta$. 1
- (ii) Write $2 \cos 2\theta \cos \theta$ as a sum of cosine ratios. 1
- (iii) Hence solve $\cos(\theta - \frac{\pi}{3}) + \cos \theta + \cos(\theta + \frac{\pi}{3}) + \cos 3\theta = 0$ for $0 \leq \theta \leq \pi$. 3

- (b) Show that: 2

$$\cot x + \frac{1}{2} \tan \frac{x}{2} = \frac{1}{2} \cot \frac{x}{2}.$$

- (c) On the number plane below, the graph of $y = \cot \frac{x}{2}$ where $-\pi \leq x \leq \pi$ has been sketched.



Copy this sketch onto your answer sheet.

- (i) On the same set of axes, sketch the graph of $y = \sin x$ where $-\pi \leq x \leq \pi$. 1
- (ii) Use the graphs and the result in part (b) to solve the inequality: 3

$$2 \cot x + \tan \frac{x}{2} - \sin x \leq 0 \quad \text{for } -\pi \leq x \leq \pi.$$

End of Examination

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(B) $-4 < x < 3$

(C) $-\frac{1}{2} < x < 4$

(D) $-3 < x < 4$

$$\begin{aligned} |2x-1| &< 7 \\ -7 &< 2x-1 < 7 \\ -6 &< 2x < 8 \\ -3 &< x < 4 \end{aligned}$$

ANSWER is D

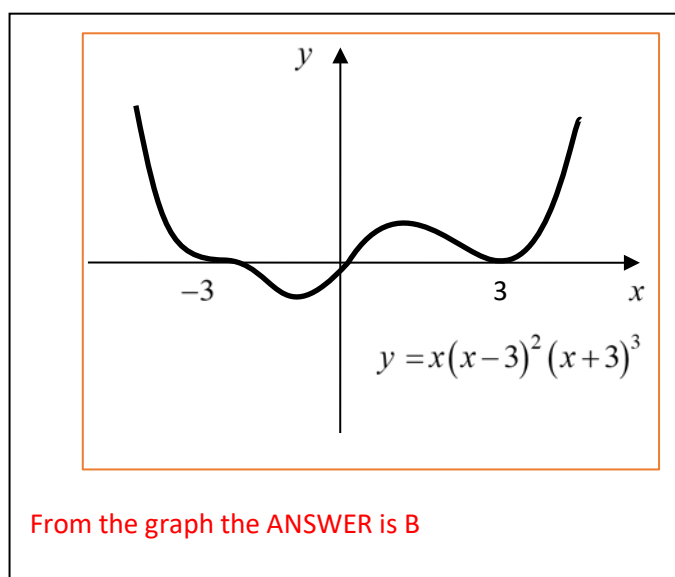
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(A) $-3 \leq x \leq 0, \quad x \geq 3$

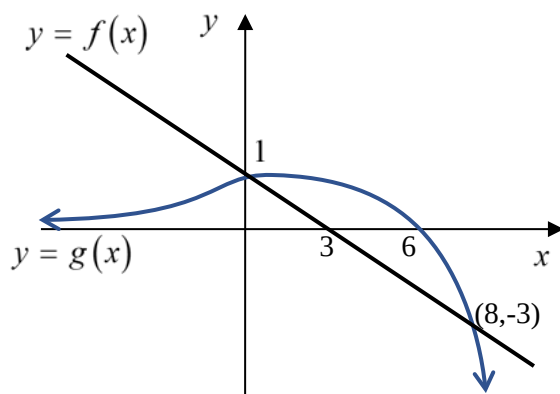
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(C) $0 \leq x \leq 3, \quad x \leq -3$

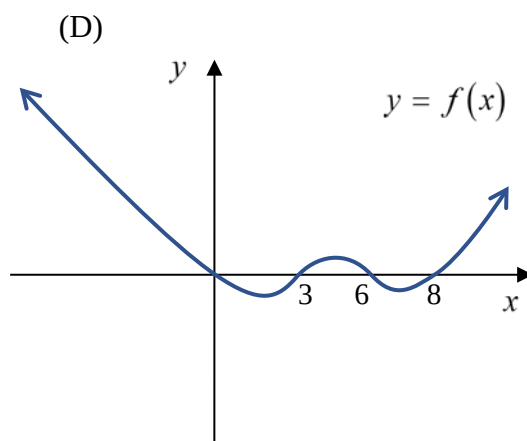
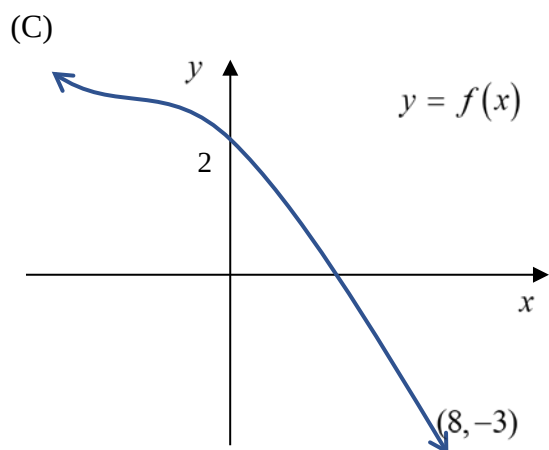
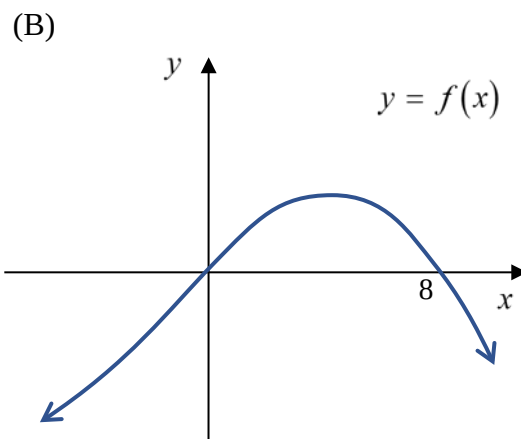
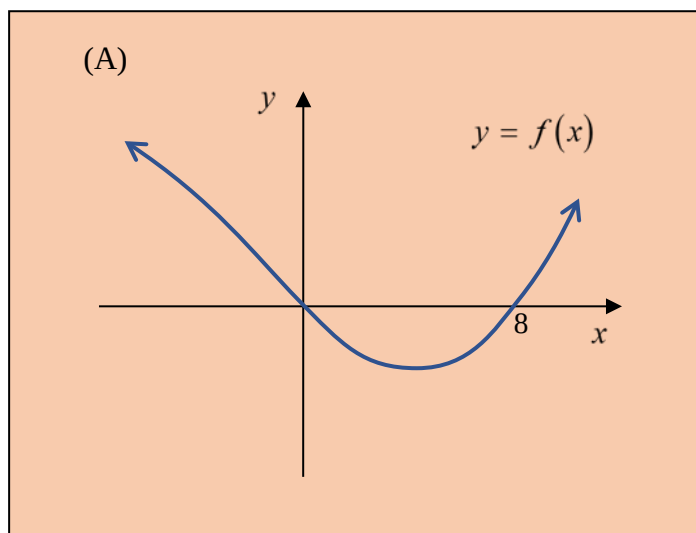
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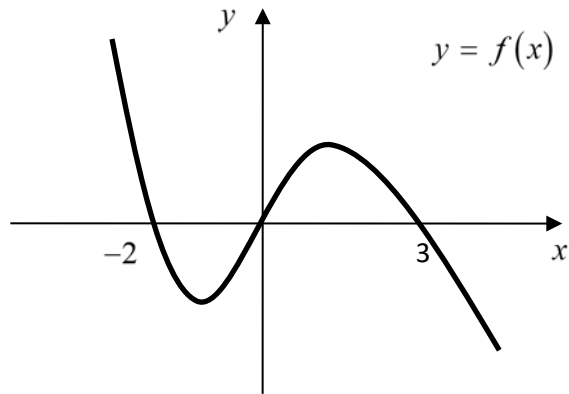
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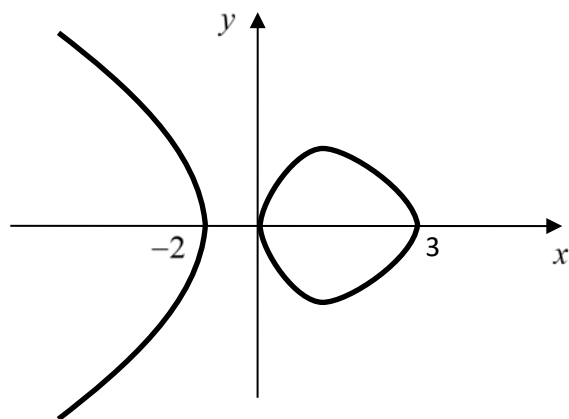
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4. The graph of the function $y = f(x)$ is shown.



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Which equation best represents the second graph?

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Question 5 (13 marks) Use a SEPARATE writing booklet.

- (a) If $f(x) = \frac{x}{x-3}$, find $g(x)$ such that $g(x) = f^{-1}(x)$ 2

$$y = \frac{x}{x-3}$$

Switch x and y and then make y the subject.

$$x = \frac{y}{y-3}$$

$$x(y-3) = y$$

$$xy - 3x = y$$

$$xy - y = 3x$$

$$y = \frac{3x}{x-1}$$

$$\therefore g(x) = \frac{3x}{x-1}$$

- (b) Solve $\frac{2x}{x-3} \leq x$. 3

$$\frac{2x}{x-3} \leq x$$

NOTE $x \neq 3$

Multiply by the square of the denominator

$$2x(x-3) \leq x(x-3)^2$$

$$0 \leq x(x-3)^2 - 2x(x-3)$$

$$0 \leq x(x-3)[(x-3)-2]$$

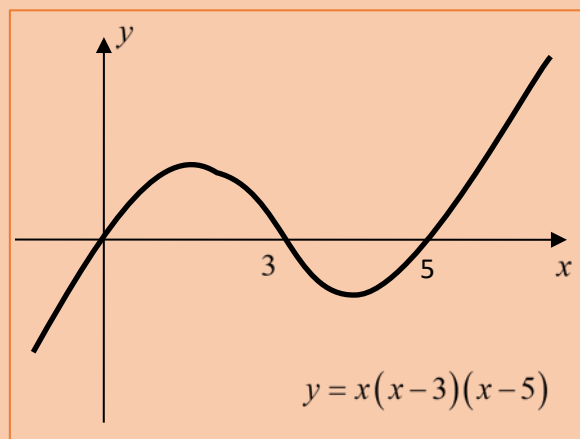
$$0 \leq x(x-3)(x-5)$$

Sketch $y = x(x-3)(x-5)$ and use the graph to solve the inequality.

From the graph, $0 \leq x(x-3)(x-5)$ when:

$0 \leq x \leq 3$ and $x \geq 5$ but since $x \neq 3$ the solution is

$0 \leq x < 3$ and $x \geq 5$



(c) By using a compound angle formula, determine the exact value of $\tan 75^\circ$.

2

$$\begin{aligned}\tan 75^\circ &= \tan(45^\circ + 30^\circ) \\&= \frac{\tan(45^\circ) + \tan(30^\circ)}{1 - \tan(45^\circ)\tan(30^\circ)} \\&= \frac{(1) + \left(\frac{1}{\sqrt{3}}\right)}{1 - (1)\left(\frac{1}{\sqrt{3}}\right)} \\&= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \\&= \frac{(\sqrt{3} + 1)^2}{3 - 1} = \frac{4 + 2\sqrt{3}}{2} = 2 + \sqrt{3}\end{aligned}$$

(d) Let $5 \cos x - \sqrt{11} \sin x \equiv R \cos(x + \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 360^\circ$.

(i) By expanding the right-hand side and equating coefficients,
show that $R = 6$ and $\alpha = 33^\circ 33'$ to the nearest minute.

3

$$\begin{aligned}5 \cos x - \sqrt{11} \sin x &= R \cos(x + \alpha) \\&= R \cos x \cos \alpha - R \sin x \sin \alpha\end{aligned}$$

By equating co-efficients of $\sin x$ and $\cos x$

$$\begin{aligned}R \cos \alpha &= 5 & I & & -R \sin \alpha &= -\sqrt{11} \\R \sin \alpha &= \sqrt{11} & II & & & \end{aligned}$$
$$I^2 + II^2$$
$$\begin{aligned}5^2 + (\sqrt{11})^2 &= R^2 \cos^2 \alpha + R^2 \sin^2 \alpha \\36 &= R^2 (\cos^2 \alpha + \sin^2 \alpha) \\6 &= R^2 \\6 &= R \quad \text{since } R > 0\end{aligned}$$

Since $\cos \alpha = \frac{5}{6}$ and $\sin \alpha = \frac{\sqrt{11}}{6}$ α must be first quadrant.

$$\alpha = \cos^{-1} \frac{5}{6} = 33^\circ 33'$$
$$5 \cos x - \sqrt{11} \sin x = 6 \cos(x + 33^\circ 33')$$

(ii) Hence, find the range of $y = 6 + 5\cos x - \sqrt{11}\sin x$.

1

$$-1 \leq \cos(x + 33^\circ 33') \leq 1$$

$$-6 \leq 6\cos(x + 33^\circ 33') \leq 6$$

$$-6 \leq 5\cos x - \sqrt{11}\sin x \leq 6$$

$$0 \leq 6 + 5\cos x - \sqrt{11}\sin x \leq 12$$

Range is $0 \leq y \leq 12$

(e) Show that $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$.

2

$$\frac{\sin 2\theta}{1 + \cos 2\theta} = \frac{2\sin \theta \cos \theta}{1 + 2\cos^2 \theta - 1}$$

$$= \frac{2\sin \theta \cos \theta}{1 + 2\cos^2 \theta - 1}$$

$$= \frac{2\sin \theta \cos \theta}{2\cos^2 \theta}$$

$$= \frac{\sin \theta}{\cos \theta}$$

$$= \tan \theta = RHS$$

End of Question 5

Question 6 (12 marks) Use a SEPARATE writing booklet.

(a) Solve $\sin 2x - \sin x = 0$ for $-\pi \leq x \leq \pi$.

3

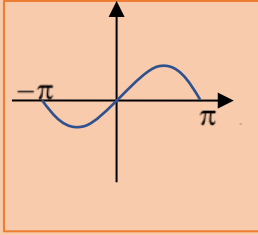
$$\sin 2x - \sin x = 0 \quad \text{for } -\pi \leq x \leq \pi.$$

$$2 \sin x \cos x - \sin x = 0$$

$$\sin x [2 \cos x - 1] = 0$$

$$\sin x = 0$$

$$2 \cos x - 1 = 0$$



$$\cos x = \frac{1}{2}$$

$$x = \pm \cos^{-1} \frac{1}{2}$$

$$x = \pm \frac{\pi}{3}$$

From the graph $x = -\pi, 0, \pi$

$$x = -\pi, -\frac{\pi}{3}, 0, \frac{\pi}{3}, \pi$$

(b) By letting $t = \tan \frac{\theta}{2}$, solve:

3

$$\frac{\cos \theta + \sin \theta - 1}{\cos \theta - \sin \theta + 1} = 3 \quad \text{for } 0 \leq \theta \leq 2\pi$$

$$\frac{\cos \theta + \sin \theta - 1}{\cos \theta - \sin \theta + 1} = 3$$

$$\frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2} - 1 = 3$$

$$\frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2} + 1 = 3$$

$$\frac{1-t^2+2t-(1+t^2)}{1-t^2-2t+(1+t^2)} = 3$$

$$\frac{2t-2t^2}{2-2t} = 3$$

$$\frac{2t(1-t)}{2(1-t)} = 3$$

$$t = 3 \quad (1-t) \neq 0$$

$$\tan \frac{\theta}{2} = 3 \quad 0 \leq \theta \leq 2\pi$$

$$\tan \frac{\theta}{2} = 3 \quad 0 \leq \frac{\theta}{2} \leq \pi$$

First Quad Only

$$\frac{\theta}{2} = \tan^{-1} 3$$

$$\frac{\theta}{2} = 1.249$$

$$\theta = 2.498$$

$$= 2.5$$

(c) Consider the parametric equations $y = \cos 2t - 1$ and $x = \cos t$.

(i) Determine the Cartesian equation in x and y .

2

$$y = \cos 2t - 1 \quad x = \cos t$$

$$y = (2 \cos^2 t - 1) - 1$$

$$y = 2 \cos^2 t - 2$$

$$y = 2x^2 - 2$$

(ii) What are the restrictions on x and y ?

2

$$-1 \leq \cos 2t \leq 1 \quad -1 \leq \cos t \leq 1$$

$$-2 \leq \cos 2t - 1 \leq 0 \quad -1 \leq x \leq 1$$

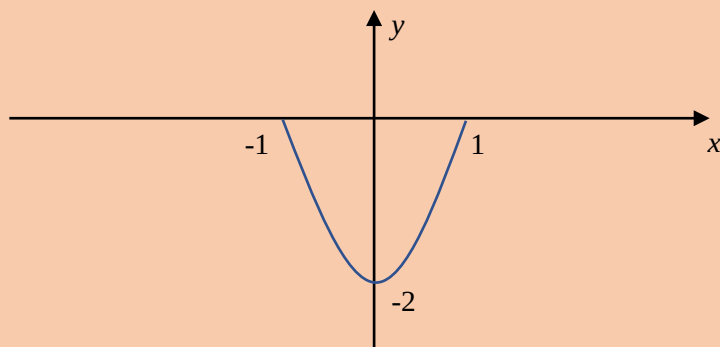
$$-2 \leq y \leq 0$$

(iii) Sketch the curve defined by $y = \cos 2t - 1$ and $x = \cos t$.

2

The Cartesian equation is $y = 2x^2 - 2$ with restrictions of

$$-2 \leq y \leq 0 \quad -1 \leq x \leq 1$$



End of Question 6

Question 7 (11 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that $\cos(\alpha - \beta) + \cos(\alpha + \beta) = 2 \cos \alpha \cos \beta$. **1**

$$\begin{aligned} LHS &= \cos(\alpha - \beta) + \cos(\alpha + \beta) \\ &= \cos \alpha \cos \beta + \sin \alpha \sin \beta + \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= 2 \cos \alpha \cos \beta \\ &= RHS \end{aligned}$$

- (ii) Write $2 \cos 2\theta \cos \theta$ as a sum of cosine ratios. **1**

$$\begin{aligned} \text{From part i) let } \alpha &= 2\theta \text{ and } \beta = \theta \\ 2 \cos 2\theta \cos \theta &= \cos(2\theta - \theta) + \cos(2\theta + \theta) \\ &= \cos \theta + \cos 3\theta \end{aligned}$$

- (iii) Hence solve $\cos(\theta - \frac{\pi}{3}) + \cos \theta + \cos(\theta + \frac{\pi}{3}) + \cos 3\theta = 0$ for $0 \leq \theta \leq \pi$. **3**

From part ii) $\cos \theta + \cos 3\theta = 2 \cos 2\theta \cos \theta$

Also part i) $\cos\left(\theta - \frac{\pi}{3}\right) + \cos\left(\theta + \frac{\pi}{3}\right) = 2 \cos \theta \cos\left(\frac{\pi}{3}\right)$

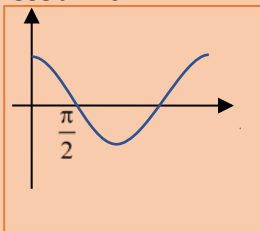
$$\cos\left(\theta - \frac{\pi}{3}\right) + \cos \theta + \cos\left(\theta + \frac{\pi}{3}\right) + \cos 3\theta = 0$$

$$2 \cos 2\theta \cos \theta + 2 \cos \theta \cos\left(\frac{\pi}{3}\right) = 0$$

$$2 \cos 2\theta \cos \theta + 2\left(\cos \theta\right)\left(\frac{1}{2}\right) = 0$$

$$\cos \theta (2 \cos 2\theta + 1) = 0$$

$$\cos \theta = 0$$



$$\cos 2\theta = -\frac{1}{2}$$

$$0 \leq 2\theta \leq 2\pi$$

$$2\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\theta = \frac{\pi}{3}, \frac{2\pi}{3}$$

$$\theta = \frac{\pi}{2}$$

$$\therefore \theta = \frac{\pi}{2}, \quad \frac{\pi}{3}, \frac{2\pi}{3}$$

(b) Show that:

2

$$\cot x + \frac{1}{2} \tan \frac{x}{2} = \frac{1}{2} \cot \frac{x}{2}.$$

$$LHS = \cot x + \frac{1}{2} \tan \frac{x}{2}$$

$$= \frac{1-t^2}{2t} + \frac{t}{2}$$

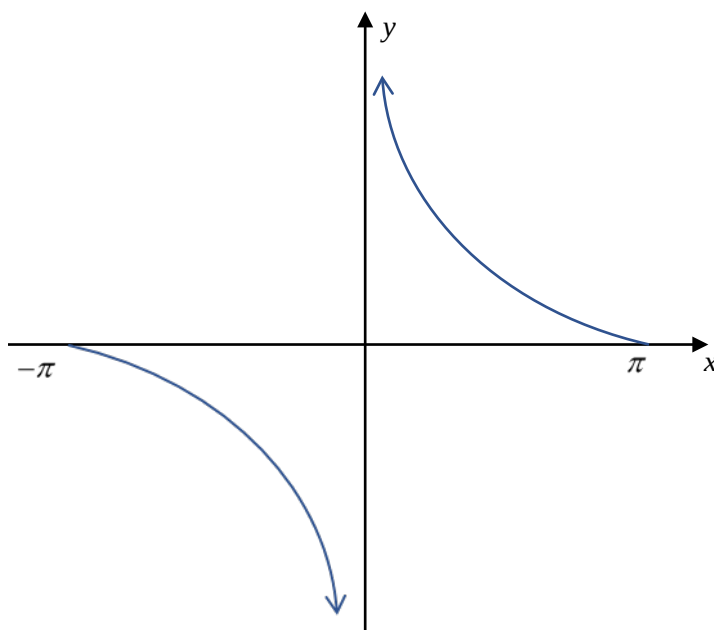
$$= \frac{1-t^2}{2t} + \frac{t^2}{2t}$$

$$= \frac{1}{2t}$$

$$= \frac{1}{2 \tan \frac{x}{2}}$$

$$= \frac{1}{2} \cot \frac{x}{2} = RHS$$

- (c) On the number plane below, the graph of $y = \cot \frac{x}{2}$ where $-\pi \leq x \leq \pi$ has been sketched.



Copy this sketch onto your answer sheet.

- (i) On the same set of axes, sketch the graph of $y = \sin x$ where $-\pi \leq x \leq \pi$.

1

Solving simultaneously

$$\cot \frac{x}{2} = \sin x$$

$$\frac{1}{t} = \frac{2t}{1+t^2}$$

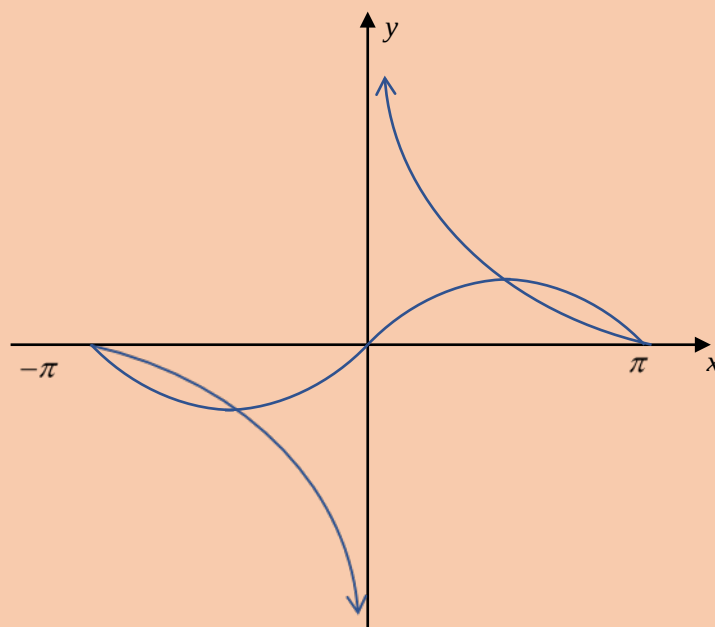
$$1+t^2 = 2t^2$$

$$t^2 = 1$$

$$\tan \frac{x}{2} = \pm 1$$

$$\frac{x}{2} = \pm \frac{\pi}{4}$$

$$x = \pm \frac{\pi}{2}$$



Graphs intersect at:

$$\left(-\frac{\pi}{2}, -1\right) \quad \text{and} \quad \left(\frac{\pi}{2}, 1\right)$$

(ii) Use the graphs and the result in part (b) to solve the inequality:

3

$$2 \cot x + \tan \frac{x}{2} - \sin x \leq 0 \quad \text{for } -\pi \leq x \leq \pi .$$

$$2 \cot x + \tan \frac{x}{2} - \sin x \leq 0$$

$$2 \cot x + \tan \frac{x}{2} \leq \sin x$$

$$2 \left(\cot x + \frac{1}{2} \tan \frac{x}{2} \right) \leq \sin x$$

$$2 \left(\frac{1}{2} \cot \frac{x}{2} \right) \leq \sin x \quad \text{From part b)}$$

$$\cot \frac{x}{2} \leq \sin x$$

Using the graph from part i) the solution is where the graph of $y = \sin x$ is above $y = \cot \frac{x}{2}$.

NOTE: $x = 0$ is not in the domain of $y = \cot x$ and

$x = \pm\pi$ is not in the domain of $y = \tan \frac{x}{2}$

So the solution to $2 \cot x + \tan \frac{x}{2} - \sin x \leq 0$ is:

$$-\frac{\pi}{2} \leq x < 0 \quad \text{and} \quad \frac{\pi}{2} \leq x < \pi$$

End of Examination

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